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Reflective Journal

This lab helped me understand how AI models are prepared for real-world use on smaller devices. Before this assignment, I only thought about training models, but I did not realize how important the conversion step is, TensorFlow Lite showed me that models must be perfected so they can run on devices with limited memory and power. This made me see how AI can be used outside of large computers and cloud systems.

One of the main challenges I faced was understanding the difference between TensorFlow and TensorFlow Lite. At first, they seemed like the same thing, but I learned that TensorFlow is for building and training models, while TensorFlow Lite is for deploying models on edge devices. Another challenging was learning how the TensorFlow Lite Interpreter works, the steps of giving tensors, setting the input tensor, invoking the model, and then reading the output felt confusing at first. I had to look carefully at the input and output details to make sure the data shape and type were correct, and making sure the test image was reshaped and converted to “float32” was also something I had to pay attention to.

Through this lab, I learned the complete pipeline of an AI model, from training to deployment. I learned how datasets like MNIST are loaded and normalized, how neural network is structured, and how training improves model accuracy. I also saw that saving a model is important because it allows the model to be reused or converted later. The most important thing I gained knowledge from though was how a trained model is converted into “.tflite” file and why that format is better for edge devices, because it is smaller, faster, and designed to run efficiently without powerful hardware.

This knowledge is especially useful in real-world situations, many devices today rely on Edge AI, such as smartphones, security cameras, medical devices, and wearable technology. Instead of sending all data to the cloud, these devices can make decisions locally using TensorFlow Lite models. For example, a smart camera could recognize numbers or objects without internet access, or a medical device could analyze patient data instantly. The process we practiced in this lab is the same process used in real systems, just on a smaller and simulated scale.

Overall, this lab made me more confident about deploying AI models, not just training them. It showed me that AI development does not end after training a model, like converting, testing, and improving models for edge devices is just as important. This experience will help me in the future projects where efficiency, speed, and real-time processing are needed.